

Tea Consumption Enhances Endothelial-Dependent Vasodilation; a Meta-Analysis



Background Tea consumption is associated with a lower risk of cardiovascular disease including stroke. Direct effects of tea components on the vasculature, particularly the endothelium, may partly explain this association. Objective We performed a meta-analysis of controlled human intervention studies on the effect of tea on flow-mediated dilation (FMD) of the brachial artery, a measurement of endothelial function, which is suggested to be associated with cardiovascular risk. Methods Human intervention studies were identified by systematic search of the databases Medline, Embase, Chemical s and Biosis through March 2009 and by hand-searching related articles. Studies were selected based on predefined criteria: intervention with tea as the sole experimental variable, placebo-controlled design, and no missing data on FMD outcome or its variability. A random effects model was used to calculate the pooled overall effect on FMD due to the intake of tea. The impact of various subject and treatment characteristics was investigated in the presence of heterogeneity. Results In total, 9 studies from different research groups were included with 15 relevant study arms. The overall absolute increase in FMD of tea vs. placebo was 2.6% of the arterial diameter (95% CI: 1.8-3.3%; P-value <0.001) for a median daily dose of 500 mL of tea (2–3 cups). This is a relative increase of approximately 40% compared to the average FMD of 6.3% measured under placebo or baseline conditions. There was significant heterogeneity between studies (P-value <0.001) that might partly be explained by the cuff position either distal or proximal to the area of FMD measurement. No indication for publication bias was found. Conclusion Moderate consumption of tea substantially enhances endothelial-dependent vasodilation. This may provide a mechanistic explanation for the reduced risk of cardiovascular events and stroke observed among tea drinkers.